Computational Analysis of 2′-Deoxyguanosine Riboswitch Folding Kinetics

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RNA folding & Novelty

Riboswitch folding is a process in which regions of an mRNA produce structures and intermediary conformations with aim to produce a smaller, more functional structure. This process can be enhanced and accelerated by the introduction of ligands into the environment. The folding of the biomolecule is novel as gene expression is governed by managing modular access to regulatory domains like ribosome binding sites and transcription terminators. The process is complex as there is energetic frustration in conformations that are not thermodynamically preferred. Structural conformation can be a result of the aptamer domain binding the ligand producing a non-local effects in which local behaviors alter global conformations. The result of this is that the folding and kinetics of folding are highly sensitive to initial conditions such as the concentrations of ions, ligands available within the environment, and temperature. This is important as riboswitches are responsible as genetic control elements.

Ionic & Ligand-mediated Environment Influencing Folding Kinetics

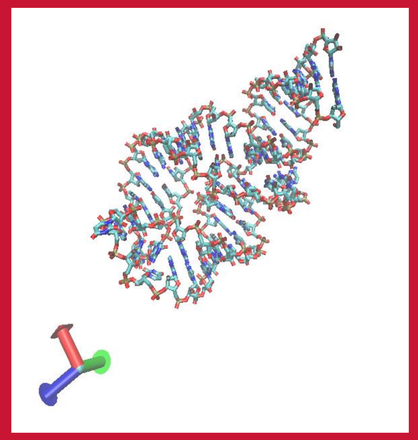
The ionic environment surrounding the mRNA is crucial to the kinetics in which folding occurs and the intermediary processes which facilitate observable non-local behaviors. RNA's have phosphate backbones which are susceptible to intermolecular forces promoting folding or express frustration. This is most crucial in simulations where ligands were not included; folding due to concentration was critical up to specific temperatures in which energetic frustration overcomes the propensity to conform to a lower energy state. Ligand binding in the RNA promotes folding as specific structural and tertiary structures stabilize and enable the folding of primary structures. Non-local interactions facilitate global folding, the introduction of ionic elements serves to accelerate and assist the folding process. This serves to highlight the importance of the functional regulation of riboswitches and the importance of the environment in which operation is optimized under.

Methodology for Analysis

To analyze RNA folding, High Performance Computing (facilitated by NOTS) operated as the modus operandi for molecular dynamics simulations. Data analytics were performed utilizing simple python scripts. SMOG and VMD were valuable tools in performing the simulations and visualizing the data. Additionally, the SMOG manual serves as a strong resource for understanding of structure based modeling and qualitative analyses to be made regarding parameters which would affect free energy and conformational propensities.

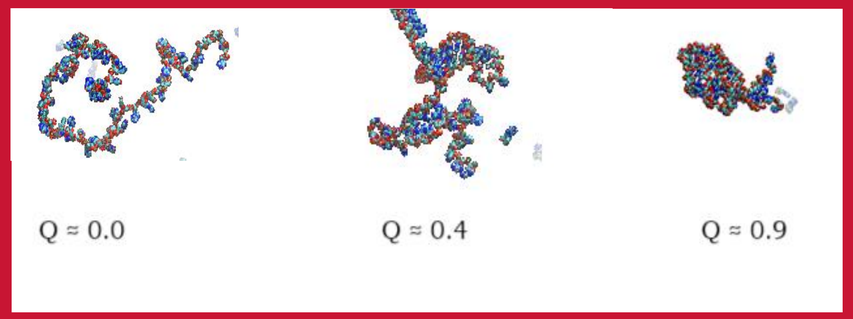
Acknowledgements:
We thank Erdong Ding for the tools & information needed to setup the simulations and environment conditions. An additional thanks is addressed to NOTS for acting as a medium for computing and for all involved with SMOG & the supplementary information provided in the manual as it was found to be helpful in the comprehension and application of scientific research. We thank the researchers of the Onuchic lab for the guidance, presentations, and environment produced in the furtherance of scientific computing of Theoretical Biological Physics. This research was conducted through the Center for Theoretical Biological Physics at Rice University. ORBITS was funded by NSF-PHY-2019745. NOTS was funded by NSF-CNS-1338099.

The images below display the native state (on the left), and an unfolded state (on the right) in the process of folding with ligands in the background and bound on the ends. The unfolded state shows the folding pathways in which central structures form before tertiary structures. The Protein Data Bank ID for the specific mRNA in presentation is "3SKI."



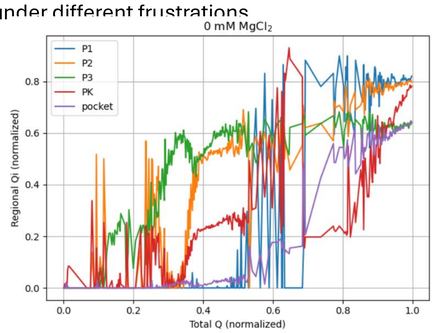


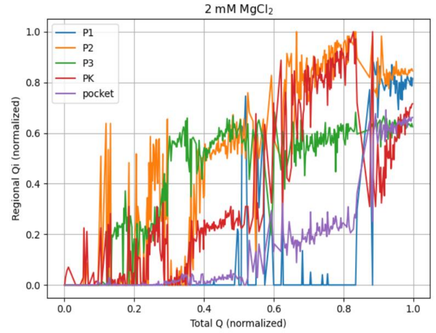
Environmental elements which alter the rate of folding include concentrations of ions, presence of ligands, and temperature. Temperature presents a drastic effect regarding the folding of the 3SKI structure and can be generalized for most, if not all biomolecules. Under conditions with ligand and concentrations even exceeding [10M] of Magnesium, increasing temperatures result in excessive time to fold and with many temperatures where folding is impermissible. The unfolded state expresses a propensity to conform to a more-stable but not folded state under isolated conditions at the temperature utilized for simulation. The image below displays an example of ranges of Q as a reaction coordinate in simulations in conformations ranging from lower, unfolded bounds of Q, to the higher Q with an almost completely folded structure.

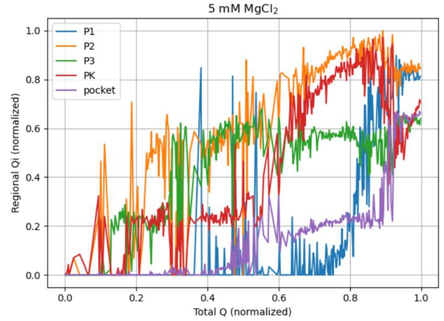


Data-Driven Folding Analytics

The images below display the regional Q vs global Q at different concentrations of magnesium ions. This shows that electrostatic interactions produce interactions which show preferential folding pathways under different frustrations.







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